AUTHORS

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Using OpenOnDemand for Real-World Interdisciplinary AI Projects

AFFILIATIONS

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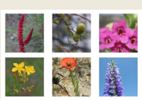
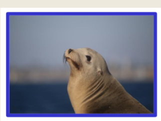
Integrating interdisciplinary collaboration in AI courses and AI research enables students to apply AI to real-world.

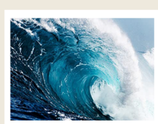
01. Introduction - Award Winning Innovative AI Teaching Methodology

Teaching Machine Learning with Applied Interdisciplinary Real World Projects

- Popular ML projects like Titanic Survival Prediction provide a great starting point, helping students learn to reuse existing code. However, they also need to be challenged with real-world data where no ready-made solutions are available online.
- Real-world projects without existing solutions help students develop creative problem-solving skills.
- An applied learning approach was implemented, supported by an internal grant and tested over two semesters.
- After starting with Google Colab and Local Jupyther Notebooks some students tried HPC platform to work more efficiently with big data.
- OpenOnDemand enabled students to handle data-intensive, interdisciplinary challenges effectively.

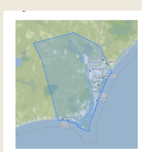


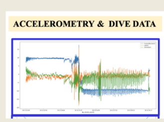


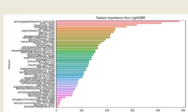


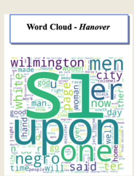
02. Objective

Demonstrate how ACCESS HPC resources, through OpenOnDemand, empower interdisciplinary real world AI projects.









03. Methodology

OpenOnDemand provides students access to GPU-based computing resources to handle large datasets and train complex models. Key techniques used include:

- **Machine Learning and Deep Learning**
- **Satellite Imagery Analysis**
- **Natural Language Processing**
- **Transformers and LLMs**
- **Time Series Analysis**

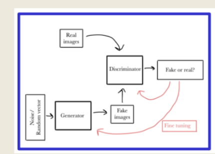
After the course, some students continued the project as their capstone in Applied AI Lab and they used OpenonDemand for faster run time.

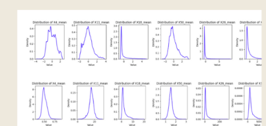
04. Key projects and Disciplines Involved

Faculty	Department	Project Topic
Dr. Ken Halanych	Earth and Ocean Sciences	Quantifying Benthos in Antarctic Images
Dr. Frederick M. Bingham	Earth and Ocean Sciences	Prediction of rainfall in western USA
Dr. Peng Gao	Earth and Ocean Sciences	Predicting Daily Firefighting Personnel Counts within the Western Conterminous United States with Deep Learning
Dr. Rachel Urbanek	Earth and Ocean Sciences	Object detection and classification in wildlife imagery
Dr. Winifred M. Johnson	Earth and Ocean Sciences	Machine Learning Analysis of Untargeted Mass Spectrometry Data to Identify Water Contaminants in Wilmington
Dr. Stacy Endriss	Earth and Ocean Sciences	Automatic Identification of Eastern North Carolina Frog Vocalizations by Machine Learning
Dr. Michel J Heijnen	Exercise Science	Understanding factors on the divers' balance using force plate data
Dr. Narcisa Pricope	Earth and Ocean Sciences	Population vulnerability vegetation classification using satellite images
Dr. Zhangping Wei	Physics and Physical Oceanography	Building damage identification using satellite images
Dr. Philip Bresnahan	Earth and Ocean Sciences	Assessment of water quality using satellite images
Dr. Michele Mendes	Nursing	Depression patterns in parents of children with special health care needs
Dr. Daniel Fisher	Healthcare Administration	Using Artificial Intelligence to Predict Patient Electronic Health Record Access Points
Dr. Michael Tift	Biology and Marine Biology	Defining the Movement of Free- Ranging Adult California Sea Lions
Dr. William Opoku-Agyeman	Healthcare Administration	Using AI to derive insights from American Hospital Association Survey Data
Dr. Jennifer Lazano	English	Analyzing a Literary Representation of the Wilmington Massacre of 1898 through NLP
Dr. Eric Moore	Environmental Sciences	Effects of Concentrated animal feeding operations (CAFO) in North Carolina Communities
Dr. Julius Jones	History	Investigating Effects of 1898 Massacre on Wilmington Black Population through AI

List of faculty that students worked with on interdisciplinary AI projects

- Students worked in teams across disciplines, guided by faculty mentors.
- Faculty mentors provided data, research problems, and domain knowledge.
- Teams began by cleaning and preparing messy, real-world data.
- Data cleaning involved handling missing fields, faulty values, and unreadable content.
- After preparation, teams applied machine learning and deep learning methods to address research questions.





05. Analysis

- OpenOnDemand improved technical skills like data handling and model training on HPC systems.
- Students learned how AI is used in different fields and the value of working with experts.
- Projects provided a scalable way to solve real-world problems.
- Students gained interest in Linux and learned how HPC platforms work.
- Working with real datasets improved data cleaning skills and prepared students for jobs.

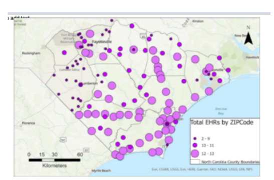
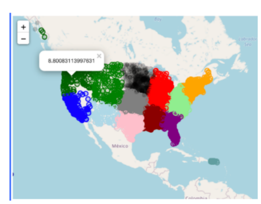
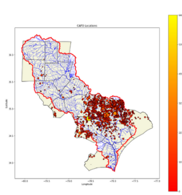


Figure 1: Total EHR by ZIPcode





06. Conclusion and Lessons Learned

Mentor Faculty and students submitted Gibbs Reflective Cycle Documents with their feedback:

- Data Cleaning Challenges: Real-world data often contains missing fields, faulty information, and misspelled text, requiring extensive cleaning efforts.
- HPC Platform Training: Short OpenDemand training videos can simplify the learning curve for students using HPC platforms.
- Mentor Collaboration: Faculty mentors without AI backgrounds focus on statistical analysis, requiring students to demonstrate the benefits of AI predictions.
- Bridging Gaps: Clear communication and framing of research problems as AI tasks improve interdisciplinary collaboration.
- Skill Development: Overcoming these challenges enhanced students' technical and problem-solving skills in real-world scenarios.